

CARNEGIE MELLON UNIVERSITY
DATA, INFERENCE & APPLIED MACHINE LEARNING (COURSE 18-785)
ASSIGNMENT 1

INSTRUCTIONS

- a. Submissions should be made via Canvas.
- b. **Single** Python/MATLAB code file(.ipynb or .m) [**Do not submit checkpoints for .ipynb**]. In addition, add comments where necessary. This demonstrates that the code is unique and owned by the student.
- c. Assignment report(.pdf) with full evidence that the assignment was completed by the student and demonstrated a full understanding of each step in the process, including textual descriptions of each result (statistics, table, graph, etc.) and insights that can be gained.
- d. Indicate the libraries you have used in your code at the beginning of the report (After the title page).
- e. Using ChatGPT for any assignment is not allowed, as it could lead to being flagged for plagiarism.

Submission process:

- a. Put source code **file and data files** in a single folder
- b. The name of the folder should be the same as your **Andrew ID**
- c. **Zip this folder and attach it on the assignment submission page (CANVAS)**
- d. After attaching the zipped file, click on "Add Another File" from the assignment submission page and **attach your report**.
- e. Submit your assignment

Specific reasons for a submission being classified as incomplete include:

- a. Failure to correctly name your folder with your Andrew ID
- b. Failure to correctly name your report, and code file with andrewID_DIAML_AssignmentNo. For example, mcsharry_DIAML_Assignment1, mcsharry_DIAML_Assignment2 and mcsharry_DIAML_Assignment3.
- c. A missing report describing the steps, results, and insights
- d. A missing dataset required for running the code
- e. A missing code file such as .ipynb or .m file
- f. An error in the file path needed to run the code

The student is responsible for checking that their submission is complete. Students will lose 10% as for late submission even if the submission is repaired during the 24 hours after the deadline has passed, and receive 0 for the assignment if it is not repaired.

The submission deadline is **Eastern Time (ET) on Monday 31st August, 2026, 17:59 PM / CAT 11: 59 PM**

No	Questions	Value
1	A piece of paper is 0.1 mm thick. However, paper thickness varies: tissue paper is around 0.05 mm, cardstock around 0.3 mm. - How many folds does it take to exceed the height of Mount Everest at 8,848 m? - How does your answer change across different paper types mentioned? - At what initial thickness would exactly 20 folds be enough? - Plot the number of folds required as a function of paper thickness, mark all 3 thickness and their folds, and explain what the shape of that curve tells you.	10%
2	The volume of water in a reservoir follows $v(t) = v(0)\exp(-at)$, where $a = 0.05$ per day. - Calculate the time for the volume to drop below 50% and 10% of its initial value. Show your working. - The exponential model implies the reservoir never reaches zero. Choose a volume threshold at which you consider the reservoir effectively empty, justify your choice, and calculate the corresponding time. - In practice, a varies with climate, season, and human activity. Repeat your calculations for $a = 0.15$ per day, plot both decay curves on the same graph, and mark the three threshold points on each. What does the difference between the two curves tell you about the reliability of predictions based on a single fixed value of a?	20%
3	You deposit \$500 in a bank account offering an annualised interest rate of 5%, compounded annually. - Calculate the balance after 1, 2, 3, 4, and 5 years. Show your working. - Look up the current savings rate offered by two real banks operating in Rwanda or East Africa. Using the same \$500 deposit, calculate the five-year balance at each bank's rate and compare it to the 5% scenario. At what deposit amount would the difference between the 5% rate and your best local bank rate exceed \$100?	10%
4	You want to buy a car worth \$20,000, tax-inclusive. A financial institution offers a loan with a monthly interest rate of 5%. - Calculate the monthly payment required to clear the debt in 1, 2, and 3 years. Round to the nearest dollar. - Rwanda is encouraging the adoption of electric vehicles by removing the 15% import tax. If you purchase an electric car of equivalent value, the effective car price is reduced by 15%. Recalculate the monthly payment for one, two, and three years under this new price. How much will you save if you pay in one, two, and three years compared to the original loan?	10%
5	You are about to set up a new business in Kigali and will invest \$100,000. On day one, you expect to have 40 customers, and the number of customers will grow at a rate of 0.5% per day. Each customer generates a profit of \$5 per day. - How many days will it take to repay your initial investment based on cumulative profits? - Plot the cumulative profits per day, mark the initial investment as a horizontal line, and mark the breakeven day clearly on the graph. - Bank of Kigali offers to double your initial investment to \$200,000 in exchange for a 50% share of daily profits. As an entrepreneur, would you take the deal if the customers grow at 0.7% due to the deal?	10%

6	Using the 2026 DRC Ebola surveillance dataset, provided on Canvas, clean the data and analyse the progression of the outbreak by answering the following. - Load the dataset and aggregate it into a single national daily series for cases and deaths. How many DRC provinces were affected? How many cases and deaths? Verify your totals against a WHO or ECDC situation report. - Check data quality: report the number of missing dates between the first and last report, at national level. Does the pattern of missing dates suggest random data loss or a structured reporting schedule? - Explain the following interpolation methods: linear interpolation, step interpolation (forward-fill), cubic spline interpolation, and exponential interpolation. Apply one method to fill the missing data to produce a clean daily series, and justify your choice. Using the cleaned series, estimate the dates when cumulative cases exceeded 100, 500, 1,000, 2,000 and 3,500, and when cumulative deaths exceeded 50, 100, 500, 1,000 and 1,500. Plot both series and mark each threshold crossing with a circle. - Calculate the average daily percentage growth rate in cases and deaths using the per-interval daily rates method and the log-linear method, and discuss the results (Which estimate would you report, and why?) - Plot cumulative deaths against cumulative cases, and discuss the case fatality rate among confirmed cases. - A colleague argues that the cleaned series you used is not real data. How would you respond?	25%
7	Using daily adjusted closing prices of SPY (equities) and TLT (long-term bonds) over the period 12/31/2021 – 08/10/2026 from Yahoo Finance: - Construct normalized price series for SPY and TLT, both starting at \$100 on 12/31/2021, and plot them on the same chart. Identify a six-month window in which both series ended lower than they started. Report the start date, end date, and the percentage change in each asset over that window. Interpret what a simultaneous decline suggests about the stability of the equity–bond diversification relationship. - Compute the 60-day rolling correlation between SPY and TLT over time. Plot the results and identify some dates where the correlation is positive. Discuss why the correlation values might be positive or negative. - Compute daily returns for both assets and summarize their empirical distribution using mean, minimum, and maximum, standard deviation, and discuss your findings.	15%